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Carbon Capture Optimization Study

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Abstract

As part of an integrated Carbon Capture, Utilization, and Storage (CCUS) development project, this study focuses on optimizing the design and operation of a CO₂ capture facility to enhance performance and reduce capture costs in hot climate environments. The facility is planned for deployment in a high-temperature desert region where ambient conditions can exceed 50°C, making cooling and energy efficiency critical design challenges. Several technical design cases were assessed using a structured, multi-dimensional framework targeting five optimization levers: improved amine solvent blends, high-pressure (HP) regeneration, absorber temperature optimization, alternative low-cost materials, and the feasibility of re-circulation of power plant exhaust gases (RoPPEG).

Solvent formulation tests demonstrated enhanced thermal stability and reduced degradation when using improved amine blends, maintaining high CO₂ capture rates. HP regeneration showed a 4.7% reduction in energy consumption compared to conventional systems. Absorber performance modeling identified an optimal design temperature range that balances cooling demand and capture efficiency, particularly under peak summer conditions. Material optimization using fiberglass reinforced plastic (FRP) showed promising capital cost savings without compromising safety or operability.

The RoPPEG strategy was evaluated through four performance scenarios at 42°C and 35% relative humidity. Results revealed a net power output reduction of ~6.5% due to reduction in CCS loads and steam requirements. At 38% RoPPEG integration, gas turbine power output dropped by 8 MW (air cooling) and 5.5 MW (hybrid cooling), primarily driven by higher return gas temperatures. However, the reduced flue gas volume enabled potential downsizing of capture plant equipment.

This paper provides novel insights into combining solvent chemistry, and equipment design for optimizing carbon capture systems in extreme climates. The study offers practical guidance for the upstream oil and gas sector aiming to accelerate CCUS deployment while maintaining economic viability and operational robustness.

Introduction

The global community is facing an acceleration of decarbonization and net-zero ambitions with a worldwide target to be reached by 2050. Carbon Capture Utilization and Storage (CCUS) will be a central technology

to reach the decarbonization goals of the global community. Petroleum Development Oman (PDO) is the leading oil and gas exploration and production company in the Sultanate of Oman. To reach its commitment to reduce its Greenhouse Gases (GHG) emissions and become a Net Zero Emission (NZE) company by 2050, PDO has identified Carbon Capture Utilization and Storage (CCUS) as a key driver. More specifically, CO₂ abatement through Enhanced Oil Recovery (EOR) has been identified as a potential double benefit solution for PDO.

Carbon Capture, Utilization, and Storage (CCUS) technologies are increasingly considered critical in reducing CO₂ emissions from large stationary sources such as gas turbines. However, cost remains a major barrier to widespread adoption, especially in regions with extreme climate conditions. Petroleum Development Oman (PDO) is pursuing a study on the application of post combustion carbon capture technology on the flue gas exhaust system at TGTU unit and Power Plant (PP).

In desert environments where ambient temperatures can exceed 50°C, conventional carbon capture systems face unique challenges related to solvent stability, cooling demand, energy efficiency, and material selection. This paper presents a comprehensive optimization study undertaken as part of a larger CCUS project to develop a capture facility that is technically viable and economically competitive under such conditions.

The scope of work includes technical assessment of CO₂ Capture technology. The following technology improvements has been studied:

1. New Amine solvents formulation that has improved capture capability
2. Re-generation of Amine at higher pressure, (HP) Regeneration to reduce the load on CO₂ compressors
3. Optimization of design through change in Ambient Design basis
4. Use of Fiberglass reinforced pipe (FRP) for Piping, Ducting, Tanks & Columns, and other lower cost material optimization
5. Recirculation of Power Plant Exhaust Gases (RoPPEG) feasibility.

This paper outlines the methodology, results, and conclusions derived from analyzing different design cases addressing these parameters.

Statement of Theory and Definitions

Carbon Capture Process: Refers to the post-combustion removal of CO₂ from flue gas streams, typically using chemical absorption via amine-based solvents.

High-Pressure Regeneration (HP Regeneration): A process in which the regeneration of solvent occurs at elevated pressures to reduce energy requirements.

Absorber Temperature Optimization: The identification of an operating temperature range that maximizes CO₂ absorption efficiency while minimizing energy consumption and cooling demand.

RoPPEG: A strategy that involves re-circulating a portion of the exhaust gas back to the turbine inlet to increase the concentration of CO₂ in the flue gas, thereby reducing the required size and cost of the capture equipment.

Fiberglass Reinforced Plastic (FRP): A composite material used as a lower-cost alternative to stainless steel or alloy in non-critical process components.

Description and Application of Equipment and Processes

CO₂ Product

CO₂ from the capture unit is injected in a compression section. The table below summarizes final CO₂ product required specifications:

Table 1—CO₂ Product Specification

Parameter	Value
CO ₂	99%+ (dry basis)
NO _x	< 10 ppmv
SO _x	< 10 ppmv
Total Sulfur	< 20 ppmv

The aim is to capture as much CO₂ as possible from flue gases. A targeted capture rate of >95% is considered.

Feed Gas

Feed gas conditions & compositions of the identified CO₂ sources were used as input to this study.

Process Description

This section describes the main units in the carbon capture plant.

1. Pre-Treatment

The aim of the pre-treatment unit is to remove contaminants (dust, Sox, etc.) from the Flue Gas before it enters the carbon capture unit. This step prevents accumulation of impurities to maintain the CO₂ capture efficiency.

The pre-treatment unit includes the following major pieces of equipment:

- Booster fans: Each emitter will be equipped with its own booster fan that increases the pressure of the flue gas from its low delivery pressure at the battery limit to compensate the pressure drop of the gas circuit.
- Caustic pre-scrubber: The objectives of the Pre-Scrubber are threefold: (1) Ensure water saturation before entering the Absorber, preventing excessive water evaporation, thus limiting water / amine make-up (2) Ensure flue-gas cooling (absorption is favored at low temperature) to the required inlet temperature of the CO₂ Absorber (3) Perform acidic components removal and ammonia reduction through a recirculating water pump around loop. For TGTU stack source, the flue gas is contaminated with SO₂ it is necessary to scrub the acid components using caustic. A 20 wt.% diluted caustic solution is used for this purpose. The caustic solution (NaOH) reacts with acidic components to form salts that are removed from the bottom of the column via a blowdown purge. The treated gas is directed to the top to undergo CO₂ absorption.

When there are no Sulphur compounds in the flue gas from power plant, only a column for direct contact cooling (DCC) will be needed to cool down the gas and saturate it with water before the absorber inlet.

2. Carbon Capture Unit

The CO₂ capture system is an amine-based technology using proprietary absorbent. The process line-up is shown in the simplified PFD below, and relies mainly on standard equipment (vessels, pumps, exchangers) and mass transfer internals (structured packing). An MVR Compressor is included in this project to reduce the duty required in the reboiler.

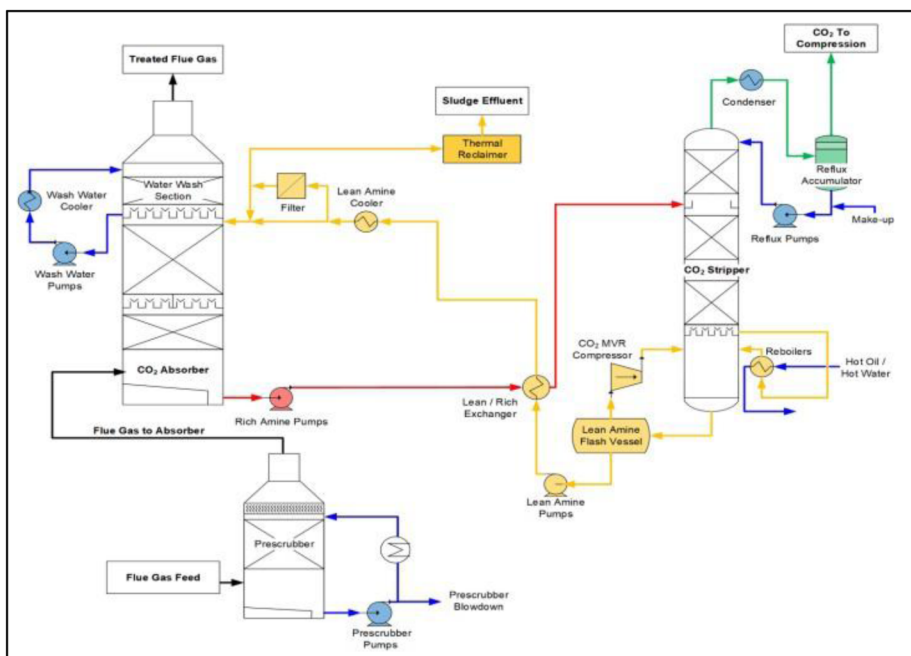


Figure 1—Standard CO₂ Capture System Simplified PFD

- Absorption: The Flue Gas enters the absorber tower from the bottom and encounters the amine solution in a counter-current flow. The absorbent forms chemical bond with the CO₂ and is absorbed from the gas stream. The rich amine, containing absorbed CO₂, is collected at the bottom of the column and pumped out to the regeneration section, while the clean gas is directed to the stack. The tower is equipped with multiple structured packing beds that ensure effective mass transfer while maintaining low pressure drop. This design enables the system to achieve a high capture efficiency of between 90% and 98%. The upper section of the absorber comprises a water-wash system that minimizes the solvent emissions and degradation products to the atmosphere.
- Water wash section: The water wash section is installed above the absorption section to capture volatile or entrained absorbent droplets and to condense water from the flue gas to maintain the water balance in the system. Wash water is circulated counter-currently to the flue gas through a packed bed in this section. Wash water is drawn from a chimney tray at the base of the Water Wash Section and is re-circulated to the top of the packed bed via the Water Wash Cooler, by the Water Wash Pump. A reflux purge from the Stripper Reflux Pump is recycled in the wash water. The Water Wash Cooler reduces the temperature of the circulating wash water, which minimizes water loss and enhances capture efficiency. Water condensed from the flue gas into the wash water section overflows from the chimney tray down into the absorption section. The treated flue gas leaving the Water Wash Section flows upwards through a mist eliminator and then is discharged to atmosphere via stack.
- Absorbent regeneration: The rich amine absorbent, loaded with CO₂, is regenerated in a stripping column, using structured packing to promote mass transfer. The column is re-boiled with Hot Oil or Hot Water to generate the stripping steam. An overhead condenser is used to condense the stripping steam from the overhead vapor circuit. The CO₂ rich gas (water saturated) is separated in the reflux accumulator and sent under pressure control to the downstream CO₂ compression section. In-between the 2 columns, a lean-rich heat exchanger is used to recover heat from the hot lean absorbent exiting the stripper column bottom and

to pre-heat the rich absorbent coming from the absorber. MVR compressor option has been considered to reduce much more the duty of the absorbent regeneration column reboiler. The stream coming from the bottom of the stripping column is flashed before entering in the lean-rich heat exchanger. The lean absorbent exiting the lean-rich exchanger is further cooled in the lean absorbent cooler and is then sent back to the Absorber column.

- Absorbent purification: The purpose of the absorbent purification section is to control the concentration of contaminants in the absorbent to maintain performance and integrity of the plant. The different elements of the purification system are:
 - Absorbent filtration: To prevent the accumulation of dusts and other insoluble contaminants, the lean amine undergoes a filtration operation consisting of a cartridge filtration system, used to remove the solids and foam promoting organics (either captured from the gas or by-products of amine degradation reactions).
 - Thermal reclaiming: The amine solution gradually accumulates stable salts and non-ionic amine degradation products that need to be eliminated from the solvent using a Thermal Reclaimer Unit. The TRU is a re-boiled column with an electrically reboiler operating under deep vacuum conditions. The degraded amine volatilizes with water allowing amine recovery in an aqueous solution at column overhead. Conversely, higher boiling impurities and heat-stable salts remain at bottom and are reclaimed as a viscous slurry for disposal.
- Absorbent Drain: Miscellaneous drains from lines and equipment containing amine are collected in the Amine Drain Tank via the Closed Amine Drains network. The drained amine can either be recycled to the CO₂ Absorber bottom (to enable regeneration of rich amine) or to the Lean Amine Tank.
- Thermal Reclaiming Unit: The absorbent accumulates both heat stable salts and non-ionic absorbent degradation products over time which must be removed from the solvent by the Thermal Reclaimer Unit (TRU). The TRU is based on a vacuum distillation process, which boils off water and amine from a slip stream of the lean absorbent and concentrates the degradation products that are then diluted in water for ease of disposal as degraded absorbent. A stream of lean absorbent is fed into the Thermal Reclaimer Tower via Lean Absorbent Pump under flow control. This stream consists of water, absorbent, heat stable salts, non-ionic degradation products and residual CO₂. 20%wt caustic solution is injected into the TRU feed line under ratio control (based on the lean absorbent flow rate) to neutralize acids and control the pH of the feed. The absorbent feed is introduced to the bottom of the thermal reclaimer column's packed section, where it is heated an electric reboiler, within the sump of the column. The thermal reclaimer column removes a purified stream of reclaimed absorbent from the column overhead. The column operates under a vacuum. The overheads are partially condensed in Thermal Reclaimer Condenser and the resulting two-phase mixture is then separated in the Thermal Reclaimer Reflux Drum. The vapor composed mostly of non-condensable components, is extracted by the Thermal Reclaimer Vacuum Unit. The liquid reflux stream from reflux drum is pumped by Thermal Reclaimer Reflux Pumps back to the top of the column on flow control, whilst the remainder of the reflux is sent to the bottom of the CO₂ stripper as reclaimed absorbent under level-to-flow cascade control in the reflux drum. The Thermal Reclaimer Bottom Pump sends column bottoms to Degraded Absorbent Tank in a batch mode. In the tank, the viscosity of degraded absorbent is reduced by addition of water from Direct Contact Cooler under a temperature-to-flow cascade control scheme. From the tank, the degraded absorbent is sent to disposal using Degraded Absorbent Pump.
- Absorbent storage: The purpose of the Lean Amine Tank is to store the fresh amine: the Makeup Amine Pump can deliver fresh amine diluted at 70 wt% (from ISO container or tanker truck)

to the Lean Amine Tank for start-up or refilling. Inline dilution of this stream is planned for with Demineralized Water (normally no flow) based on ratio flow control to reach 50% weight amine in the storage tank. The Lean Amine Tank then feeds the CO₂ Absorber with lean amine make-up, which is delivered by the Amine Feed Pumps under flow control.

- o Emissions and Effluents Gaseous Effluents: The absorbent, as well as its main degradation products, have a low volatility. As a result, a single stage water-wash water at the top of absorber is sufficient to limit emissions of these components well below ppm level. Emissions of ammonia also remain low and are of the order of ppm level or below for most applications. Nitrosamines are present at low concentration in the absorbent and have low volatility, they are typically emitted at ppb level or below. VOCs are also very low. Aldehydes emissions are also very low are typically below 0.1 ppm for acetaldehyde and not detectable for formaldehyde. Process Vents During normal operation there are no process vents to the flare or atmosphere from the Shell CO₂ Capture unit. Liquid Effluents The liquid effluents from the capture system are generated by the quench/direct contact cooler and the Thermal Reclaimer. The Thermal Reclaimer bottoms are a concentrated stream composed of the degradation products separated from the amine in the TRU distillation column. They are normally diluted to 50%wt water to facilitate storage and transport and are typically disposed of offsite as hazardous material.

RoPPEG

Exhaust gas recirculation is a technique used to reintroduce a portion of the exhaust gas into the gas turbine's inlet after appropriate cleaning and conditioning. The RoPPEG flow is mixed with the fresh ambient air upstream compressor inlet.

The CO₂ concentrations in exhaust gas from various sources vary greatly, which has an impact on the performance, design, and choice of with carbon capture system (CCS) technologies as well as capital expenditure (CapEx) and operational expenditure (OpEx). The exhaust gas from industrial land-based gas turbines contains between 3%mol-4.5%mol CO₂ at near- atmospheric pressure, depending on the gas turbine frame (E, F, or HA class machine). The low pressure and density of flue gas result in substantial volume flows, necessitating large equipment and ducts.

These factors, in turn, are reflected in the plant footprint and total installed cost. Furthermore, a lower CO₂ concentration leads to higher CO₂ capture energy requirements. The minimum work required to separate CO₂ from a gas mixture can be calculated using the first and second laws of thermodynamics and applying Gibbs free energy. The calculations revealed that as the concentration of CO₂ decreases, the minimum work required for separation increases. Therefore, compared to sources with high CO₂ concentration, removing CO₂ from sources with low CO₂ concentration, such as gas turbines is more difficult and energy intensive. Therefore, exhaust gas recirculation is a practical approach to increase CO₂ concentration and improve separation efficiency. The RoPPEG ratio is defined as the ratio of the returned flue gas to gas turbine to stack exhaust flow. RoPPEG offers two key advantages. First, the directed flue gas to the carbon capture plant is reduced proportionally with the RoPPEG rate. This reduction in mass flow to the CCS decreases the size of major CCS components, including the DCC and the absorber, leading to a decrease in both footprint and CapEx of the CCS. Secondly, since RoPPEG increases the concentration of CO₂ in the exhaust gases, resulting in a higher CO₂ partial pressure in the flue gas the driving force in the separation process is enhanced. In other words, RoPPEG allows the CCS to capture an equivalent amount of CO₂ from exhaust gases while treating a smaller volume of flue gases. This leads to a reduction in the CCS footprint, CapEx, and OpEx.

Furthermore, RoPPEG helps mitigate amine degradation linked to O₂ in the exhaust gas. This reduction in amine degradation leads to a decrease in amine make-up flow, thereby improving the overall OpEx of the CCS.

As the RoPPEG ratio increases, the O₂ concentration to the gas turbine combustor decreases, and this limitation is crucial to avoid challenges in combustion stability and dynamics. GE has been exploring RoPPEG for decades (historically studied for NO_x benefits) and has more than 20 patents in this space.

The RoPPEG flow needs to be cooled down before being directed back to the gas turbine inlet and mixed with ambient air. The RoPPEG cooling loop typically has its dedicated DCC, configured to achieve cleaning and cooling targets to ensure safe gas turbine operation. The RoPPEG DCC cooler is configured as a packed quench column, where cooling water is recirculated to the top of the column and sprayed in a counter-current direction. The RoPPEG gas flow meets the spraying water through the packing, where the RoPPEG flow is cooled down, and a portion of its water vapor condenses. Determining the RoPPEG cooling temperature involves a trade-off between gas turbine performance and RoPPEG cooler costs. Depending on the site location and water availability, the RoPPEG cooling loop can be configured with a water cooler, air cooler, hybrid cooler or chiller system. After the exhaust flow exits the RoPPEG direct contact cooler, it continues toward the inlet mixer.

The images below are a simplified ‘typical’ representation of an integrated NGCC / CCS plant without RoPPEG and with RoPPEG.

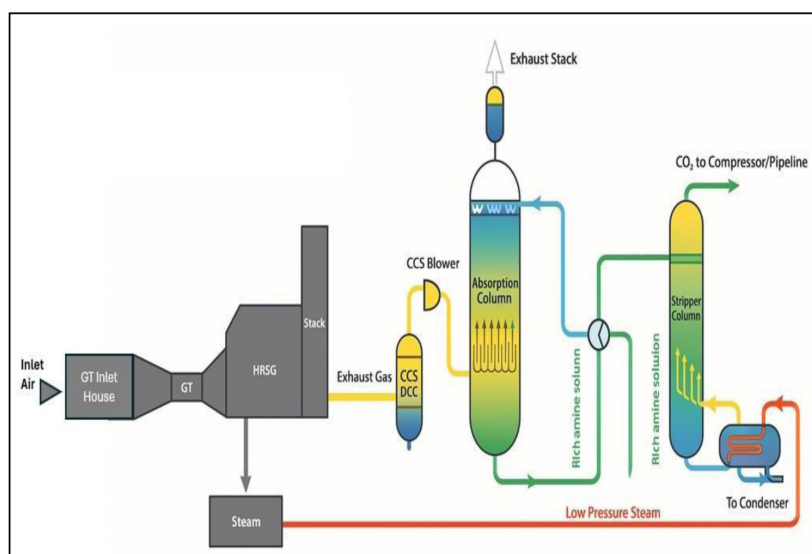


Figure 2—NGCC / CCS plant without RoPPEG

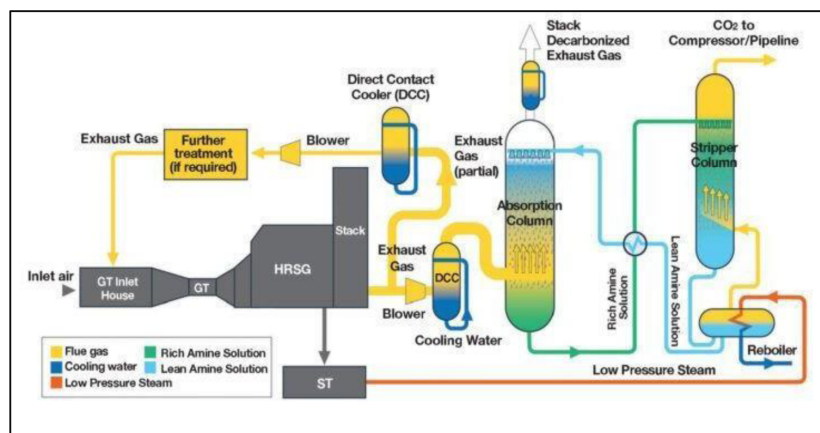


Figure 3—NGCC / CCS plant with RoPPEG

The RoPPEG loop consists of direct contact cooler (DCC), a proprietary enhanced cleaning system, water cooling loop, blower, dampers, control system and RoPPEG inlet mixer.

Presentation of Data and Results

Solvent Performance

Petroleum Development Oman (PDO) decided to study the application of post combustion CO₂ carbon capture technology on two locations. The emission sources at both the locations are as follows:

- The vent stack of the TGTU SO₂ unit
- Exhaust of the Power Plant downstream of the HRSG, and

As part of the study, simulations of the CO₂ capture unit were conducted for the different cases studied using the latest advanced solvent. High level Heat & mass balance, equipment summary sheets of the CO₂ capture unit and associated utility summaries were prepared.

Improved amine blends were evaluated for absorption efficiency, degradation rates, and thermal stability. Improved blends-maintained 90%+ capture efficiency. Degradation was reduced by 20–30% compared to conventional solvent.

HP Regeneration

A high-pressure stripper was simulated to determine energy savings versus a conventional low-pressure unit. Steam demand, reboiler duty, and operational feasibility were assessed.

Operating the regenerator at higher pressure has the advantage of producing the CO₂ product stream at a higher pressure which saves on re-compression power cost and may reduce the number of compression stages required. However, steam consumption could be reduced by operating regenerator at lower pressure. With higher pressure, MP steam due to the higher Regeneration temperature. If they regenerated at a lower pressure, then LP steam would be acceptable. Operating the Regenerator at higher pressure leads to an increase in solvent make-up rates.

HP regeneration considered with Regenerator Overhead pressure of 4 bara. Achieved 4.7% energy savings. Reduced reboiler duty and boiler steam requirements.

Absorber Temperature Optimization

The facilities shall be designed for local ambient conditions of the Sultanate of Oman. Maximum summer temperatures can peak up to 50 degC. During winter and the night, the ambient temperature is lower, and air coolers will be able to achieve lower temperatures. It may be worth designing for lower ambient temperatures and accepting that the CO₂ recovery may be lower during warm ambient conditions.

Air cooling only is considered following evaluations as the cooling medium. No chilled water / mechanical refrigeration is considered. Capture unit designed for 95% CO₂ capture rate for summer conditions. Capture rate in winter conditions is expected to be marginally higher (~95.5% for TGTU SO₂ Vent Stack and ~96.8% for PP)

The CO₂ capture unit may be optimized to capture slightly less CO₂ at peak summer temperatures, in favor of an overall smaller design, that can average 93 to 95% CO₂ capture rate year-round. The loss of 1-2% of CO₂ capture (year average) could result in over 10% reduction in some portions of the capture facility cost.

A range of absorber inlet temperatures (30–55°C) was modeled to understand the trade-offs between capture efficiency and cooling duty, especially during peak summer operations. Optimal performance observed at 40–45°C inlet temperature. Cooling load reduced by 12% without compromising CO₂ capture.

Process outlet temperature of air coolers can be considered at 53°C instead of 59°C to optimize process conditions, except during extreme summer conditions. The assessment of TGTU SO₂ exhaust gas (30/39°C) results increase in CO₂ Capture rate to ~95.5% with the hardware considered (50/59°C). Due to low feed gas temperature (38.7 °C) and high CO₂ content in the feed gas (~20 mol%), maximum utilization of solvent capacity is feasible even at the higher lean solvent temperature thereby leading to less impact of lower solvent temperature on overall capture rates. However, the impact of lower ambient temperatures on Capture rates can be seen significantly in Power Plant case due lower CO₂ feed concentration and higher feed gas temperature.

Temperature Selection: Ambient / Lean solvent to absorber temperature ranges from 50°C / 59°C to 41°C / 50°C were evaluated initially. However, based on the below justifications, optimum temperature for designing is considered as 50°C / 59°C TGTU SO₂ Vent Stack and 44°C / 53°C power plant.

- TGTU SO₂ Vent Stack: Ambient temperature of 50°C (or 59°C lean solvent temperature to absorber) considered. Ambient temperature sensitivity reflected minimal impact on equipment design as solvent loading is maximized due to:
 1. low feed gas temperature,
 2. enough driving force due to high CO₂ % in feed gas
- Power Plant: Ambient temperature of 44°C (or 53°C lean solvent temperature) considered as it resulted in substantial impact on equipment sizing within the capture unit as compared to a conservative ambient temperature of 50°C (or 59°C lean solvent temperature)

FRP Material of Construction

Equipment including tanks, ducting, and atmospheric piping were evaluated for FRP substitution. Cost, durability, and chemical compatibility were reviewed against process requirements. Capital cost savings estimated at 15–25% for applicable components. No significant impact on reliability or process compatibility.

For cases with the Vent Stack as emission source, there is a corrosion issue on flue gas equipment and lines. This problem comes from the presence of SO₂/SO₃ combined with the presence of water and a low temperature leading to condensation and formation of sulfuric acid. As gas duct has an important size for all those cases (between 1.4 and 3.0 m), to reduce cost, FRP (Fiberglass Reinforced Plastic) instead of SS304L or SS316L can be considered.

Nevertheless, considering climatic conditions of this project, it is important to highlight that FRP is sensitive to UV rays, leading to yellowing, loss of transparency, brittle pipes with microcracks, loss of cohesion between resin and glass fibers, loss of mechanical resistance leading to a brittle surface. A good solution is to add an anti-UV additive, painting (polyurethane) or coating. To improve durability, pieces exposed to external conditions can have an anti-UV varnish or gelcoat.

RoPPEG Scenarios

The study presents the exhaust gas recirculation (RoPPEG) as a technique for recirculating portion of the flue gas back to gas turbine inlet. Recirculating portion of the gas turbine exhaust gas back to the gas turbine inlet increases CO₂ concentration which improves carbon capture plant capture rate. Furthermore, RoPPEG reduces the flue gas directed to the carbon capture plant which consequently reduces the size of the carbon capture plant and footprint.

The study examined RoPPEG performance at the power plant's ambient temperature of 42°C, using a fin-fan cooler as an RoPPEG cooling loop system. The identified RoPPEG ratio for the combined cycle is 38%. The study presents high-level plant performance. The study looked at three cases that represented the power plant's performance before and after carbon capture plant integration, as well as with RoPPEG. In addition, the plant's performance was evaluated using the gas turbine Advanced path upgrade AGP.

Table below shows the combined cycle power plant conditions. These conditions are the base of the performance calculations.

Table 2—Power Plant Conditions

Ambient Temperature	oC	37
Relative Humidity	%	35
Atmospheric Pressure	mbar	991
Gas Turbine Exhaust Conditions		
Ar	Vol. %	0.88
CO ₂	Vol. %	3.79
H ₂ O	Vol. %	9.23
N ₂	Vol. %	73.65
O ₂	Vol. %	12.45
Exhaust Gas Temperature	oC	606.4
Exhaust Gas Flow	t/h	674.8

The site current emissions CO₂, NO_x and CO are listed in Table below.

Table 3—GT Current Emissions

CO ₂ Emissions	One GT
Ton/hr	47.4
MTPA	0.4
Emissions	Max Limits ppmv
NO _x	25
CO	9

The current study presents four scenarios. Case 1 represents the current NGCC performance. Case 2 is the plant integrated with a carbon capture plant, with all required steam for CCS supplied by an auxiliary boiler. Cases 3 and 3a is plant integration with a carbon capture plant and 38% RoPPEG, with steam supplied entirely from the auxiliary boiler. The study also analyzed the cases with gas turbine AGP upgrade.

Table 4—The Study Cases

Case	CCS	Amb. Temp (oC)	RoPPEG Cooling	RoPPEG	Steam Source
1	OFF	42			
2	ON	42	-	-	Auxiliary Boiler
3	ON	42	Air Cooling	38%	Auxiliary Boiler
3a	ON	42	Hybrid Cooling	38%	Auxiliary Boiler

Thoughtful integration of the new carbon capture plant into the existing combined cycle plant is imperative to ensure optimal performance of the combined equipment operating as one adept site. RoPPEG modeling was done using Promax. A thorough mass and heat balance for the plant and cases presented. The power plant and carbon capture plant and RoPPEG block diagram is shown in Figure below.

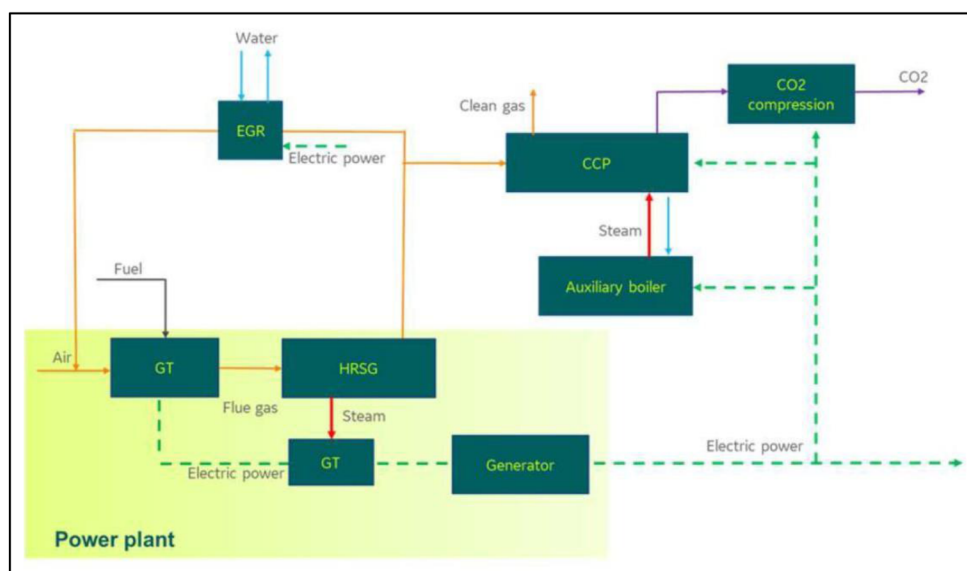


Figure 4—RoPPEG Integration

Due to water scarcity at the location, the selected RoPPEG cooling loop is an air-cooled system. The study evaluated the performance of RoPPEGs under design ambient temperature 42°C. The efficiency of the air-cooling system is directly influenced by the dry ambient conditions. The cooling efficiency of the air-cooling system is significantly decreased as the ambient temperature rises, resulting in a significant decrease in system performance.

- Air cooling system using fin-fan cooler: The air fin-fan cooler is located directly in the RoPPEG DCC cooling loop where the ambient air directly cooling the circulating water discharged in the bottom of the RoPPEG DCC. The selected pinch point between the cooling water temperature at the outlet of the air fin-fan cooler and dry bulb ambient temperature is 10°C. Due to the corrosivity of the water in the RoPPEG loop, the air fin-fan cooler needs to be made of stainless steel SS316. The air fin-fan cooler has cooling limitations at high ambient temperatures which limits its effectiveness. When the circulating water temperature in the RoPPEG loop approaches the ambient temperature, the cooling efficiency of the air-cooler drops. The temperature of the circulating water will thus exceed the dew point temperature of the flue gas. As a result, the RoPPEG does not work as efficiently in clearing pollutants from the exhaust, and the flue gas will not undergo condensation.
- Hybrid chiller (air fin-fan cooler + mechanical chiller): A viable option is to incorporate a hybrid chiller into the RoPPEG loop, specifically designed to operate just at elevated ambient temperatures. The hybrid chiller functions by first chilling the RoPPEG water circulation utilizing free cooling via the air fin fan cooler and then providing extra cooling through the mechanical chiller. This will decrease the workload on the chiller and allow the RoPPEG to operate efficiently, even in hot weather conditions. The installation of a mechanical chiller raises the cost of RoPPEG owing to the added expenses of the equipment and the installation for the chiller. In addition, the site experiences an increase in parasitic load when the mechanical chiller is operating. This will provide an indication of the capital expenditure (CapEx) increase resulting from the installation

of a chiller. A centrifugal refrigeration unit, more appropriate for the RoPPEG application, was chosen as the mechanical chiller. This is because of its industrial sizes which more appropriate for RoPPEG application.

The RoPPEG cooling loop has been modeled for ambient temperatures of 42°C and 50°C, and with a hybrid chiller and fin-fan.

Table 5—RoPPEG cooling loop performance results

		Fin fan cooler	Hybrid cooler	Fin fan cooler	Hybrid cooler
Assumptions					
Amb. Temp	oC	42	42	50	50
RoPPEG Ratio	%	38	38	38	38
Cooler TDD	oC	10	10	10	10
L/G ratio	-	2.7	2.6	2.7	
Results					
RoPPEG Flow	Kg/s	64.3	64.3	64.3	64.3
Water mass flow	Kg/s	174	167	174	167
Water condensation	Kg/s	0.056	2.47	0	2.47
Water makeup	Kg/s	0	0	1.8	0
RoPPEG return temp.	oC	54	44	59	44
RoPPEG return water content	%mol	14	8.64	17.8	8.63

The key performance indications of the carbon capture performance are the CO₂ capture rate, CO₂ purity, CO₂ delivered pressure to in the stripper and the steam to CO₂ captured ratio. The study is targeting 95% capture rate from both the gas turbines and auxiliary boiler.

The energetic of the carbon capture plant differs depending on the selected carbon capture technology provider. The conventional carbon capture plant using MEA as solvent showed a specific reboiler duty SRD around 4-5 GJ/tonneCO₂. The commercial state of the art carbon capture plant which utilizing a blend solvent shows much lower values. In this study specific reboiler duty of 3.06 used as a basis. The modeling has showed that carbon capture plant performance improves by 7% when using RoPPEG,

Table 6—Specific Reboiler Duty

Conditions	Specific reboiler duty (GJ/tonCO ₂)
Carbon capture plant	3.06
Carbon capture plant with 38% RoPPEG	2.8

The performance has been evaluated at ambient temperature 42°C and 35% relative humidity. The case performance has also been evaluated with a gas turbine AGP upgrade. The results demonstrate that integrating the carbon capture plant with the power plant reduces the plant net output by around 6.5% due to

the power required to run the CCS and CO₂ compression train. Due to the CCS load and the steam required from the auxiliary boiler for CO₂ regeneration, the site net efficiency, which includes the fuel consumption of the auxiliary boiler, decreases by 14.6%. Cases 3 and 3a show the plant's performance at 38% RoPPEG with air cooling and hybrid cooling. The RoPPEG return temperature is approximately 54°C with a fin-fan cooler and 44°C when a hybrid cooler is employed. The gas turbine power drops by approximately 8MW with an air-cooling system and 5.5MW with a hybrid cooling system due to increased RoPPEG return temperatures. The RoPPEG value of reducing CCS CapEx and OpEx will be explored in the study's next phase, which will include a techno-economic analysis of the scenarios.

Conclusions

This study identified several cost and energy optimization levers applicable to CO₂ capture facilities in hot climates. Improved solvent blends and HP regeneration contribute to lower energy consumption, while absorber temperature tuning provides a practical approach for cooling optimization. Material substitution using FRP is a viable cost-reduction strategy for non-critical equipment. RoPPEG implementation, while reducing turbine output slightly, can significantly reduce flue gas volume and CCS equipment size.

Combined, these strategies support the economic and technical viability of deploying carbon capture systems in harsh environments typical of upstream oil and gas operations.

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Nomenclature

AGP	= Advanced Gas Path
CCS	= Carbon Capture and Storage
CO ₂	= Carbon Dioxide
FRP	= Fiberglass Reinforced Plastic
HP	= High Pressure
MEA	= Monoethanolamine
RH	= Relative Humidity
RoPPEG	= Recirculation of Power Plant Exhaust Gases
TGTU	: Tail Gas Treatment Unit